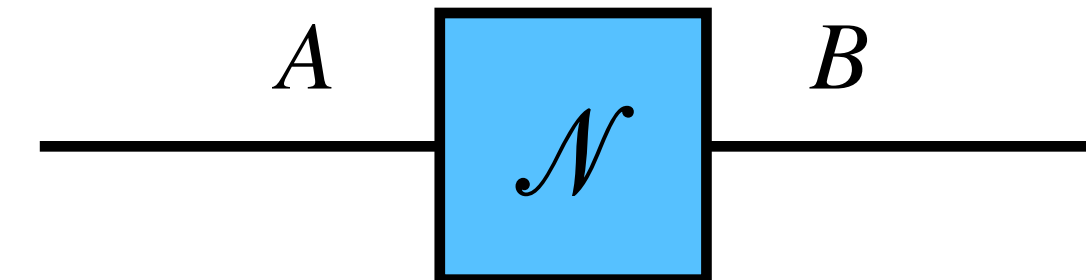


Postselected communication over quantum channels

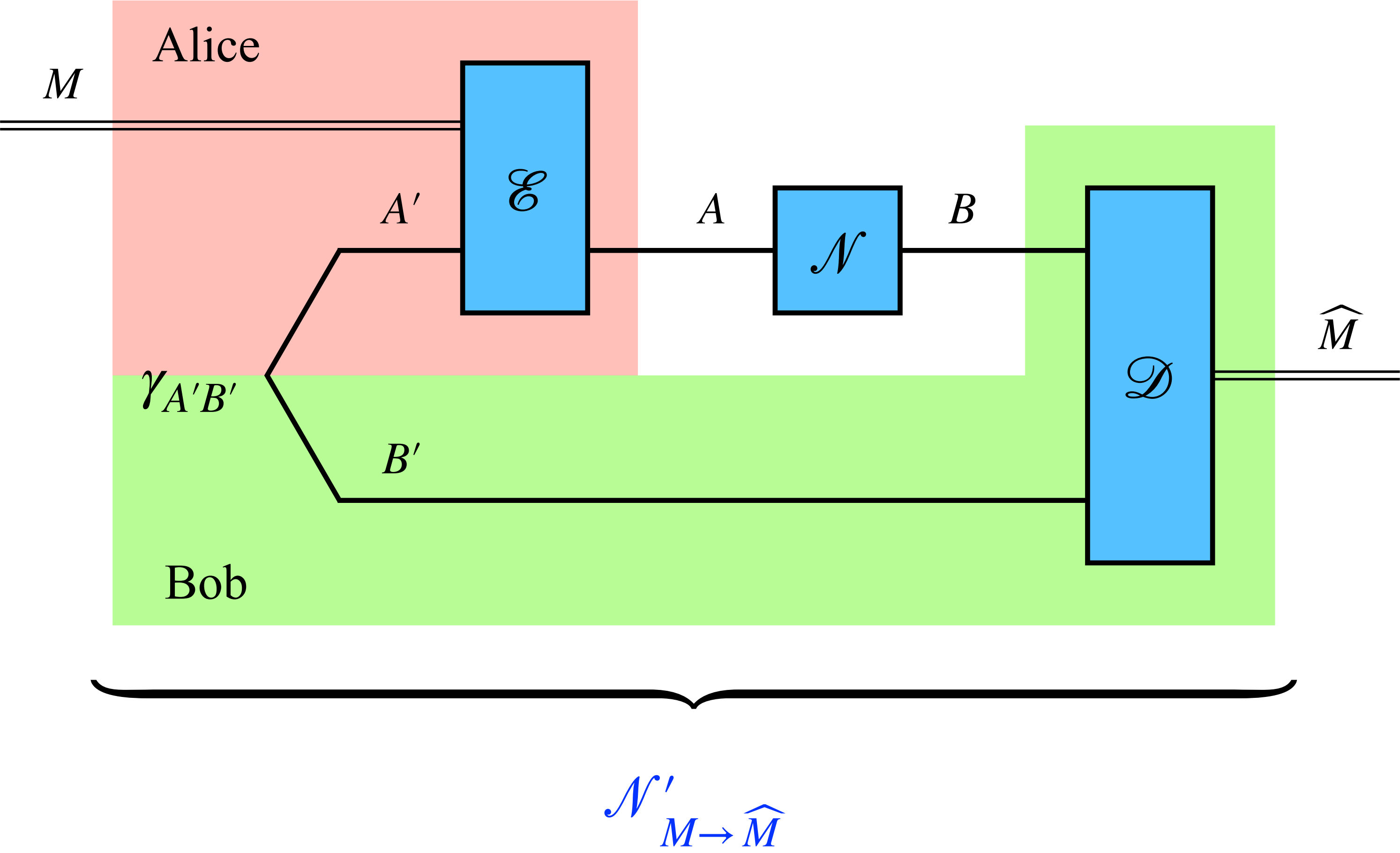
Kaiyuan Ji, Bartosz Regula, and Mark M. Wilde

arXiv:2308.02583

Entanglement-assisted (EA) classical communication



Entanglement-assisted (EA) classical communication



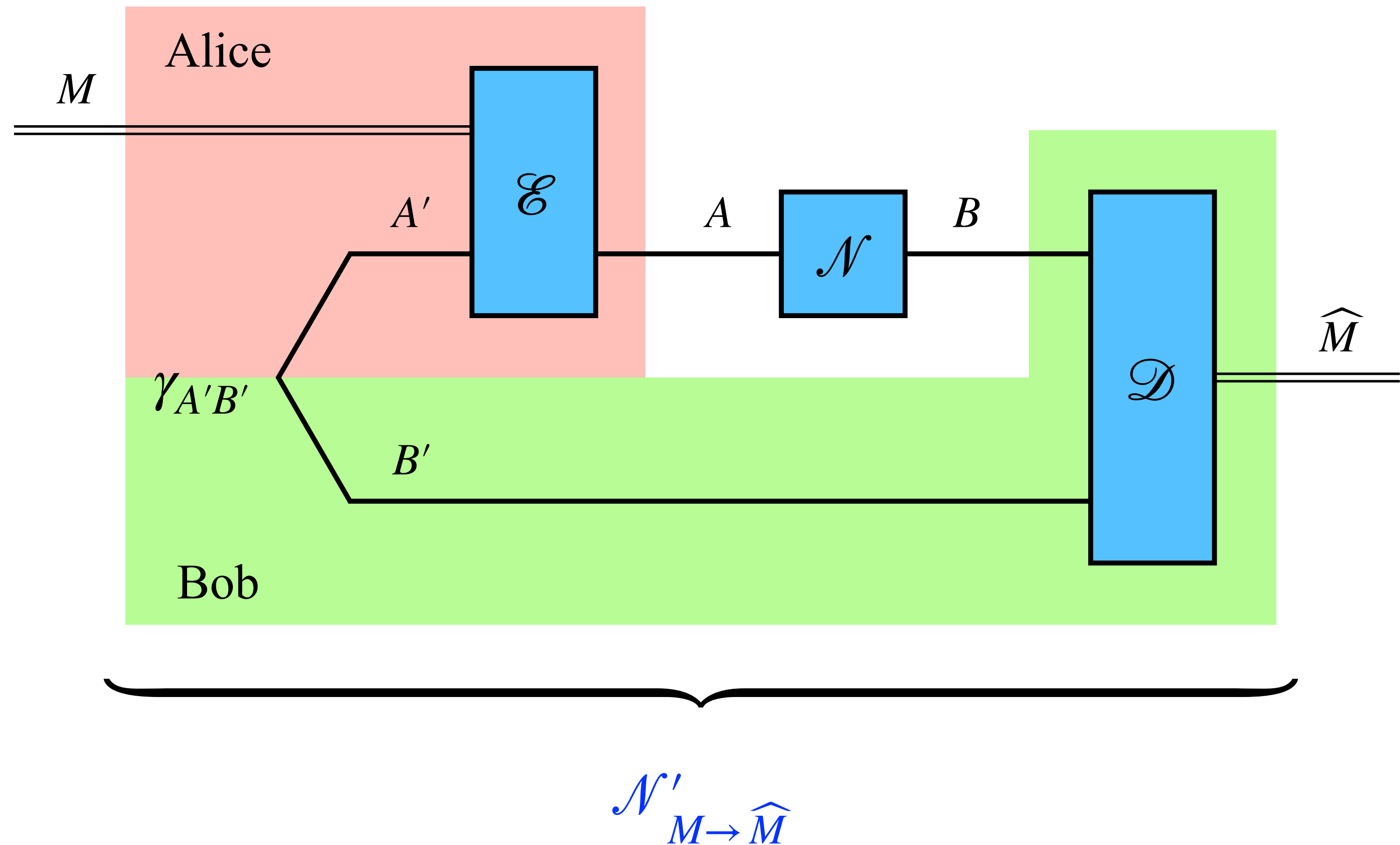
Entanglement-assisted (EA) classical communication

Given a quantum channel $\mathcal{N}_{A \rightarrow B}$,

- Worst-case error probability:

$$P_{\text{err}}(\mathcal{N}')$$

$$:= \max_{m \in [d_M]} \Pr \left\{ \hat{M} \neq m \mid M = m \right\}.$$



Entanglement-assisted (EA) classical communication

Given a quantum channel $\mathcal{N}_{A \rightarrow B}$,

- One-shot EA classical capacity:

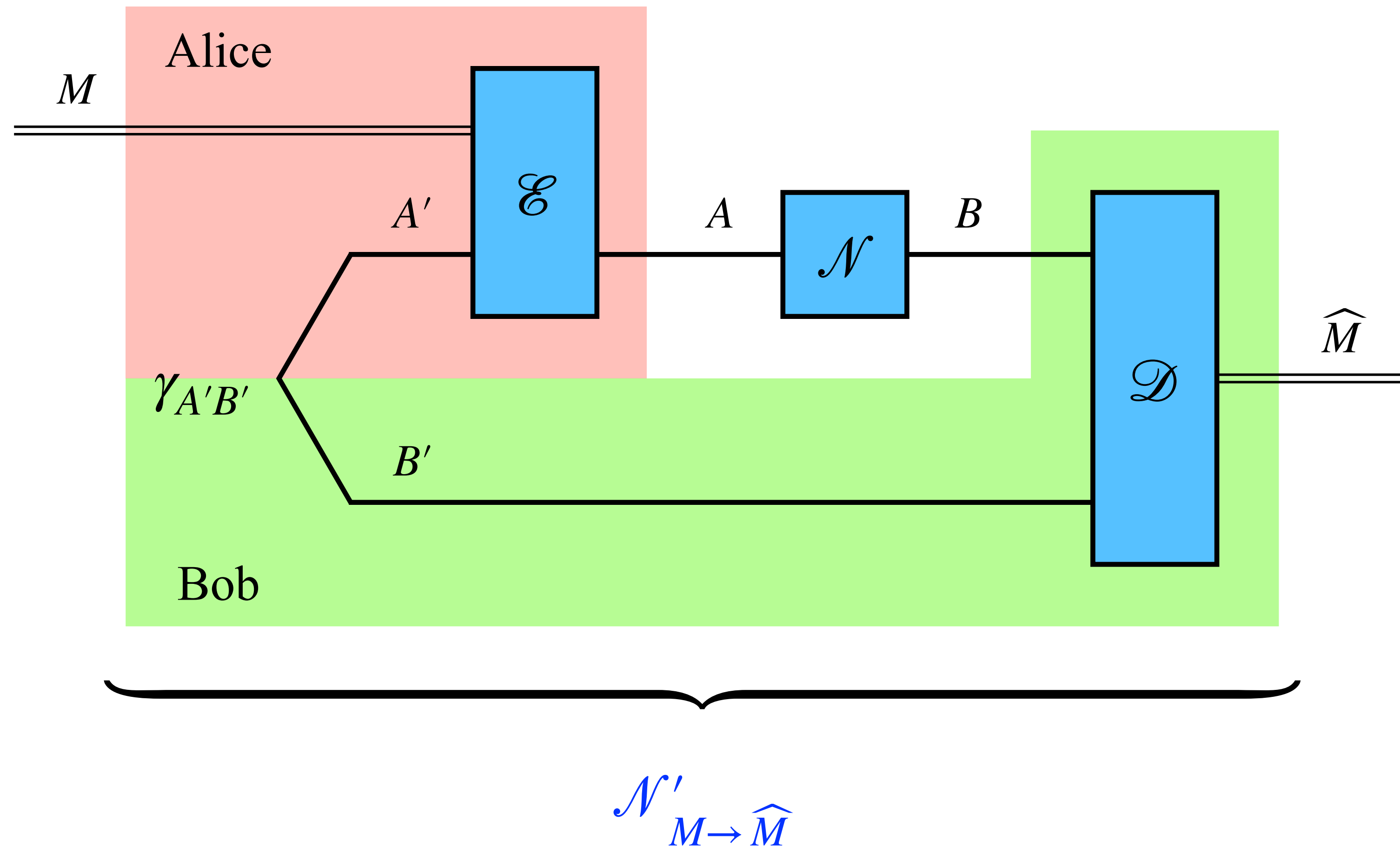
$$C_{\text{EA}}^\varepsilon(\mathcal{N}) :=$$

$$\sup_{\mathcal{E}, \mathcal{D}, \gamma} \left\{ \log_2 d_M : P_{\text{err}}(\mathcal{N}') \leq \varepsilon \right\}.$$

- Asymptotic EA classical capacity:

$$C_{\text{EA}}(\mathcal{N}) :=$$

$$\inf_{\varepsilon \in (0,1)} \liminf_{n \rightarrow \infty} \frac{1}{n} C_{\text{EA}}^\varepsilon(\mathcal{N}^{\otimes n}).$$



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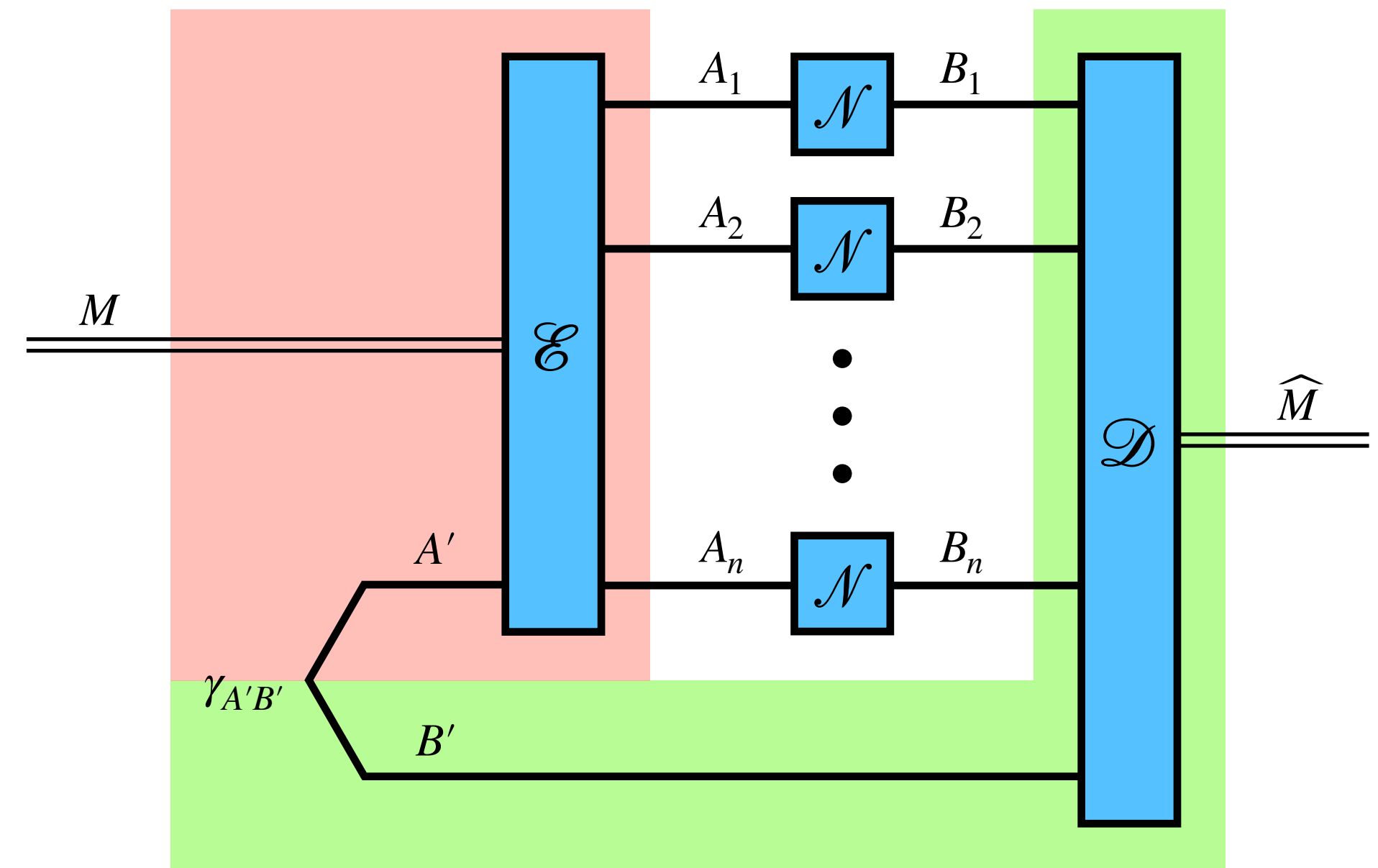
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Entanglement-assisted (EA) classical communication

Entanglement-assisted capacity theorem:

$$C_{\text{EA}}(\mathcal{N}) = I(\mathcal{N}),$$

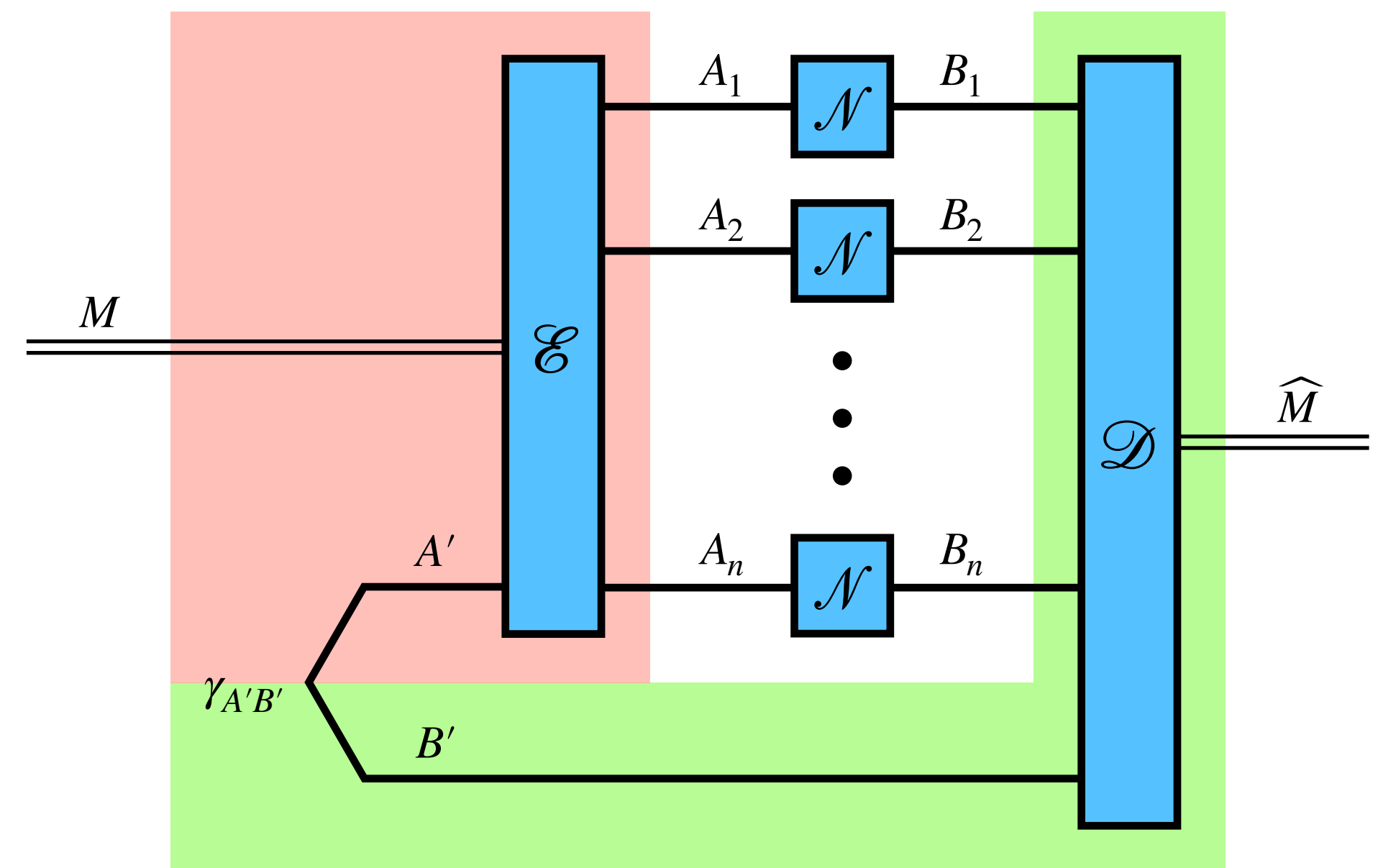
where

$$I(\mathcal{N}) := \sup_{\rho_{RA}} I(R; B)_{\mathcal{N}_{A \rightarrow B}[\rho_{RA}]},$$

$$I(R; B)_{\omega_{RB}} := \inf_{\sigma_B} D(\omega_{RB} \| \omega_R \otimes \sigma_B).$$

(Bennett et al., 2002)

(Holevo, 2002)



“Postselected” communication

If Bob has the superpower of doing **postselection** when decoding the message, how does this superpower affect the fundamental limit of communication?

Postselection in real life

- Postselection in a nutshell: not making attempts due to fear of failure
- Poker games: “fold” on bad private cards
- Coursework: drop a course if performed poorly in mid-term
- Sports: shoot easy shots, pass tough shots to teammates

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vs.



“Postselected” classical communication

In conventional communication:

- Given Alice’s message $M = m \in [d_M]$, Bob’s guess should be $\hat{M} \in [d_M]$.
- If $\hat{M} = m$, it’s a “hit;” otherwise it’s a “miss.”

“Postselected” classical communication

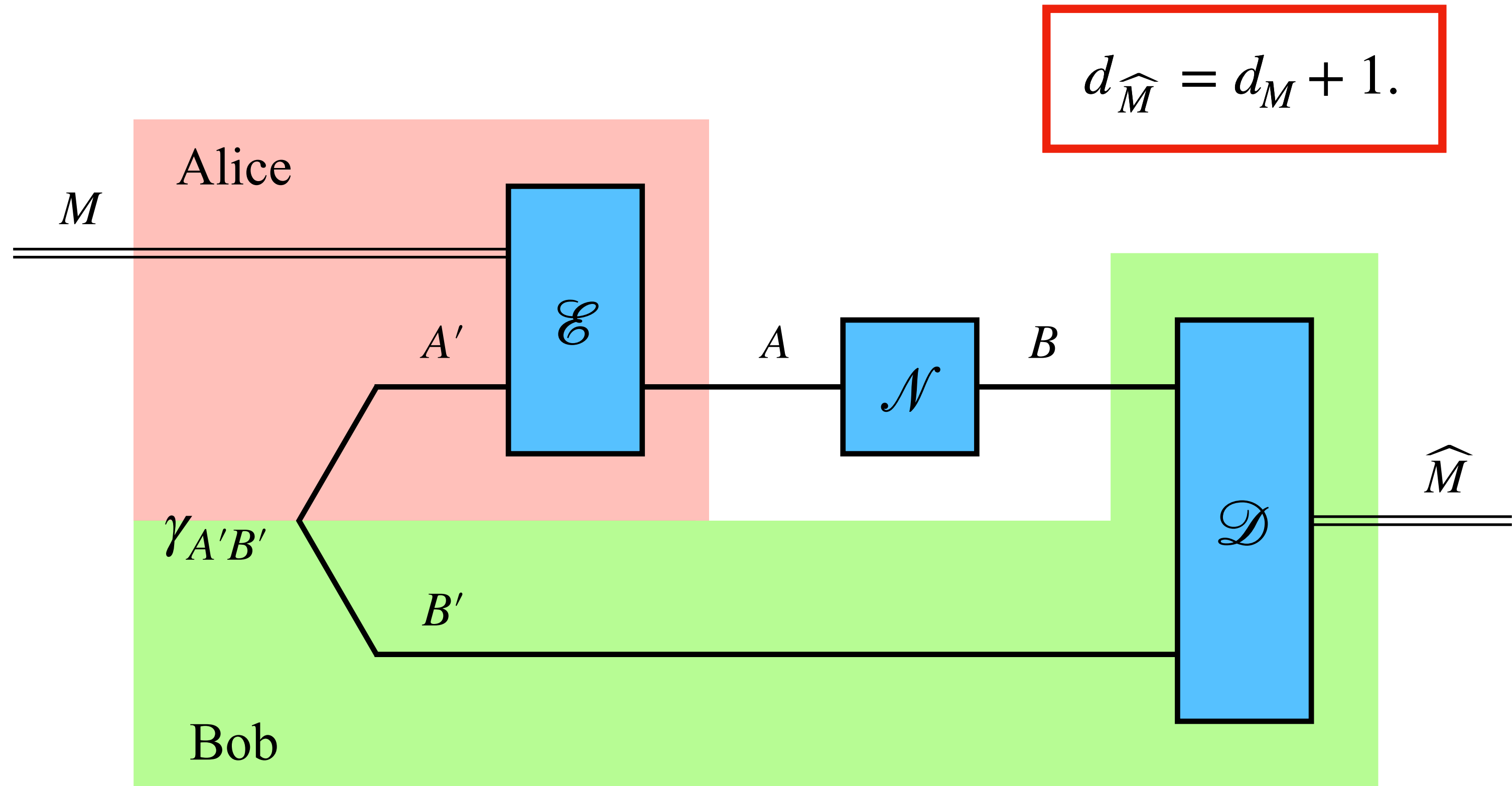
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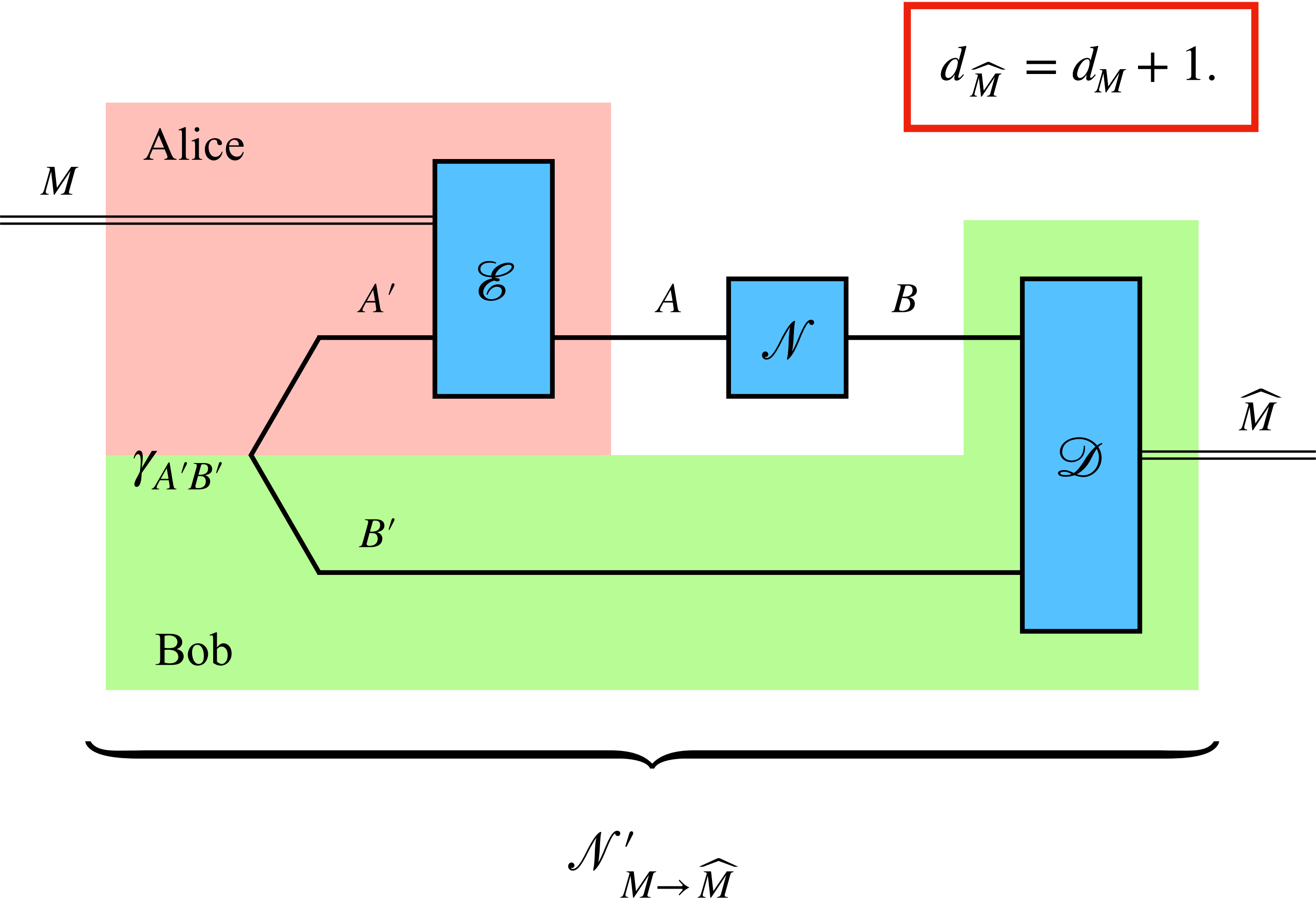
In **postselected** communication:

- Given Alice’s message $M = m \in [d_M]$, Bob’s output can be $\hat{M} \in [d_M] \cup \{ \perp \}$.
- If $\hat{M} \in [d_M]$: If $\hat{M} = m$, it’s a “hit;” if $\hat{M} \neq m$, it’s a “miss.”
- If $\hat{M} = \perp$, this means Bob refuses to make a guess (i.e., Bob is “**inconclusive**”).
- Error rate = (# misses) / (# hits + # misses). Inconclusive trials are **dismissed**.
- In other words, the goal is now to **simulate an erasure channel**.

Postselected worst-case error probability

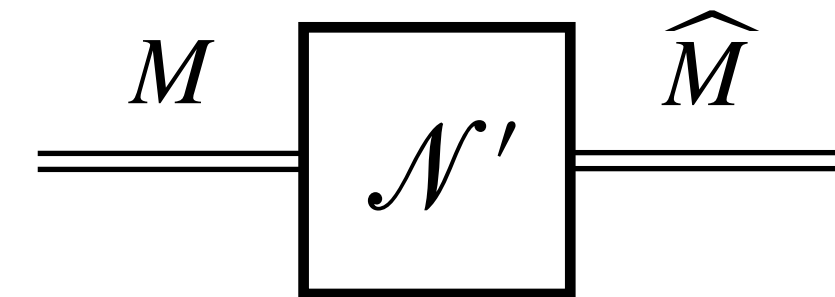


Postselected worst-case error probability



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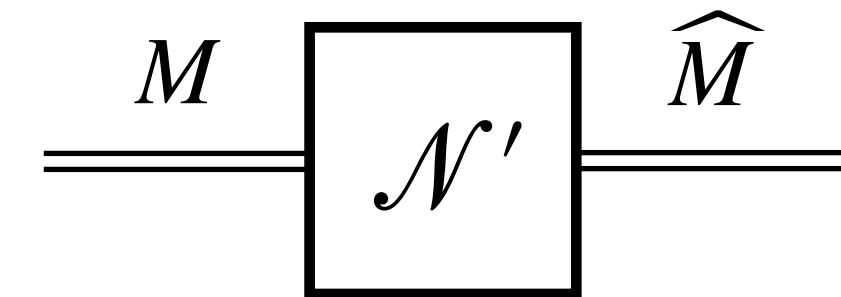
$$d_{\hat{M}} = d_M + 1.$$



Postselected worst-case error probability

$$d_{\widehat{M}} = d_M + 1.$$

$$\begin{aligned} P_{\text{err}}^*(\mathcal{N}') &:= 1 - \min_{m \in [d_M]} \Pr_{\widehat{M}} \left\{ \widehat{M} = m \mid M = m, \widehat{M} \in [d_M] \right\} \\ &= 1 - \min_{m \in [d_M]} \frac{\Pr_{\widehat{M}} \left\{ \widehat{M} = m \mid M = m \right\}}{\sum_{m' \in [d_M]} \Pr_{\widehat{M}} \left\{ \widehat{M} = m' \mid M = m \right\}} \\ &= 1 - \min_{m \in [d_M]} \frac{\langle m | \mathcal{N}' [|m\rangle\langle m|] | m \rangle}{\sum_{m' \in [d_M]} \langle m' | \mathcal{N}' [|m\rangle\langle m|] | m' \rangle}. \end{aligned}$$



Postselected entanglement-assisted (pEA) capacity

Given a **quantum** channel $\mathcal{N}_{A \rightarrow B}$,

- One-shot pEA classical capacity:

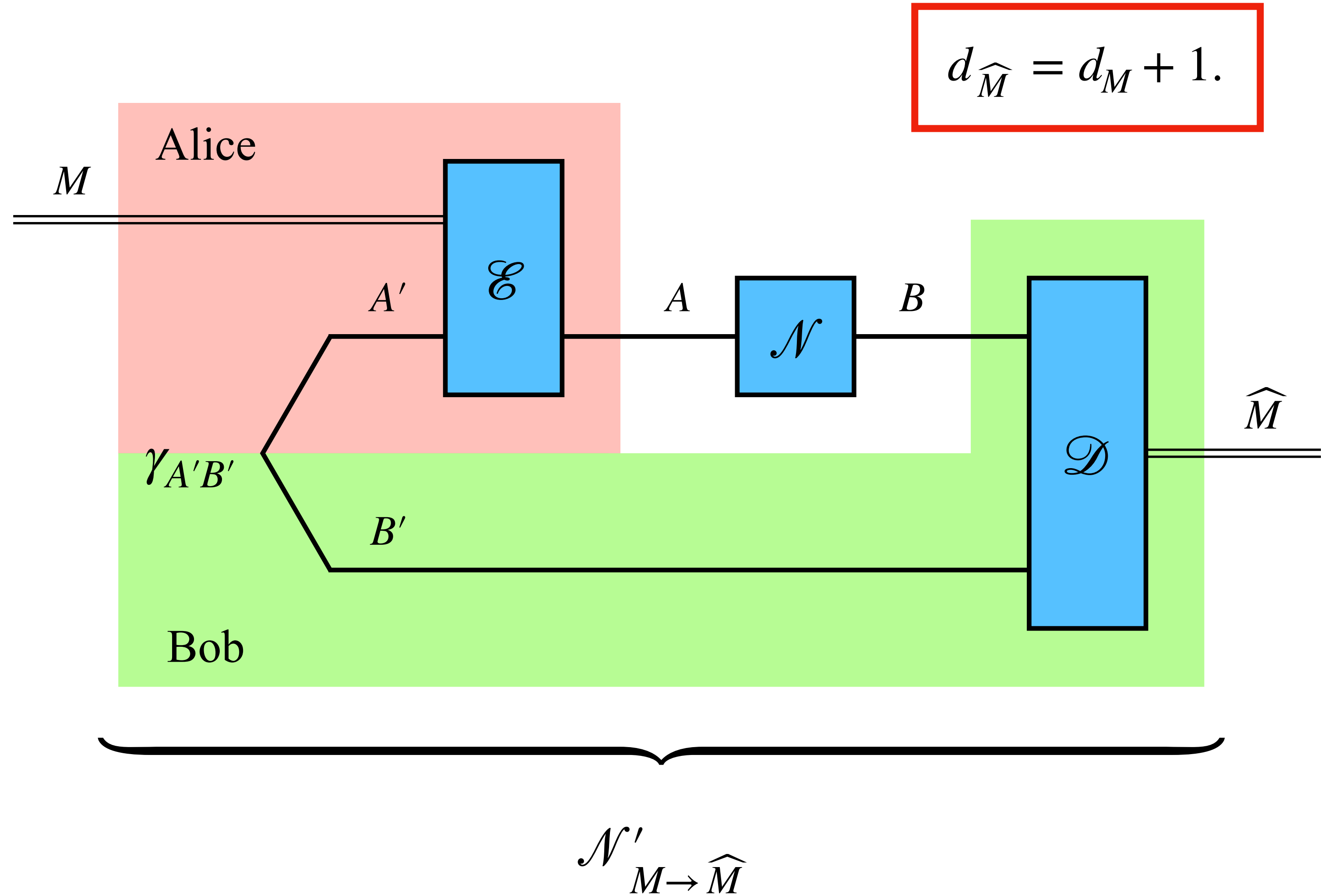
$$C_{\text{pEA}}^\varepsilon(\mathcal{N}) :=$$

$$\sup_{\mathcal{E}, \mathcal{D}, \gamma} \left\{ \log_2 d_M : P_{\text{err}}^*(\mathcal{N}') \leq \varepsilon \right\}.$$

- Asymptotic pEA classical capacity:

$$C_{\text{pEA}}(\mathcal{N}) :=$$

$$\inf_{\varepsilon \in (0,1)} \liminf_{n \rightarrow \infty} \frac{1}{n} C_{\text{pEA}}^\varepsilon(\mathcal{N}^{\otimes n}).$$



Main result: Characterisation of pEA capacities

One-shot pEA classical capacity:

$$C_{\text{pEA}}^{\varepsilon}(\mathcal{N}) \approx \log_2 \left(\frac{\varepsilon}{1 - \varepsilon} 2^{I_{\Omega}(\mathcal{N})} + 1 \right).$$

Asymptotic pEA classical capacity:

$$C_{\text{pEA}}(\mathcal{N}) = I_{\Omega}(\mathcal{N}).$$

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Here:

$$I_{\Omega}(\mathcal{N}) := \sup_{\rho_{RA}} I_{\Omega}(R; B)_{\mathcal{N}_{A \rightarrow B}[\rho_{RA}]},$$

$$I_{\Omega}(R; B)_{\omega_{RB}} := \sup_{\sigma_B} D_{\Omega}(\omega_{RB} \| \omega_R \otimes \sigma_B),$$

$$D_{\Omega}(\rho \| \sigma) := D_{\max}(\rho \| \sigma) + D_{\max}(\sigma \| \rho).$$

(Bushell, 1973)
(Reeb et al., 2011)

$I_{\Omega}(\mathcal{N})$ is SDP-computable.

Proof sketch: Converse

Prove that

$$C_{\text{pEA}}^\varepsilon(\mathcal{N}) \leq \sup_{\rho_{RA}} \inf_{\sigma_B} D_{\text{pH}}^\varepsilon(\mathcal{N}_{A \rightarrow B} [\rho_{RA}] \parallel \rho_R \otimes \sigma_B),$$

where $D_{\text{pH}}^\varepsilon(\rho \parallel \sigma)$ is the postselected hypothesis testing relative entropy.

Proof sketch: Converse

(Regula et al., arXiv:2209.10550)

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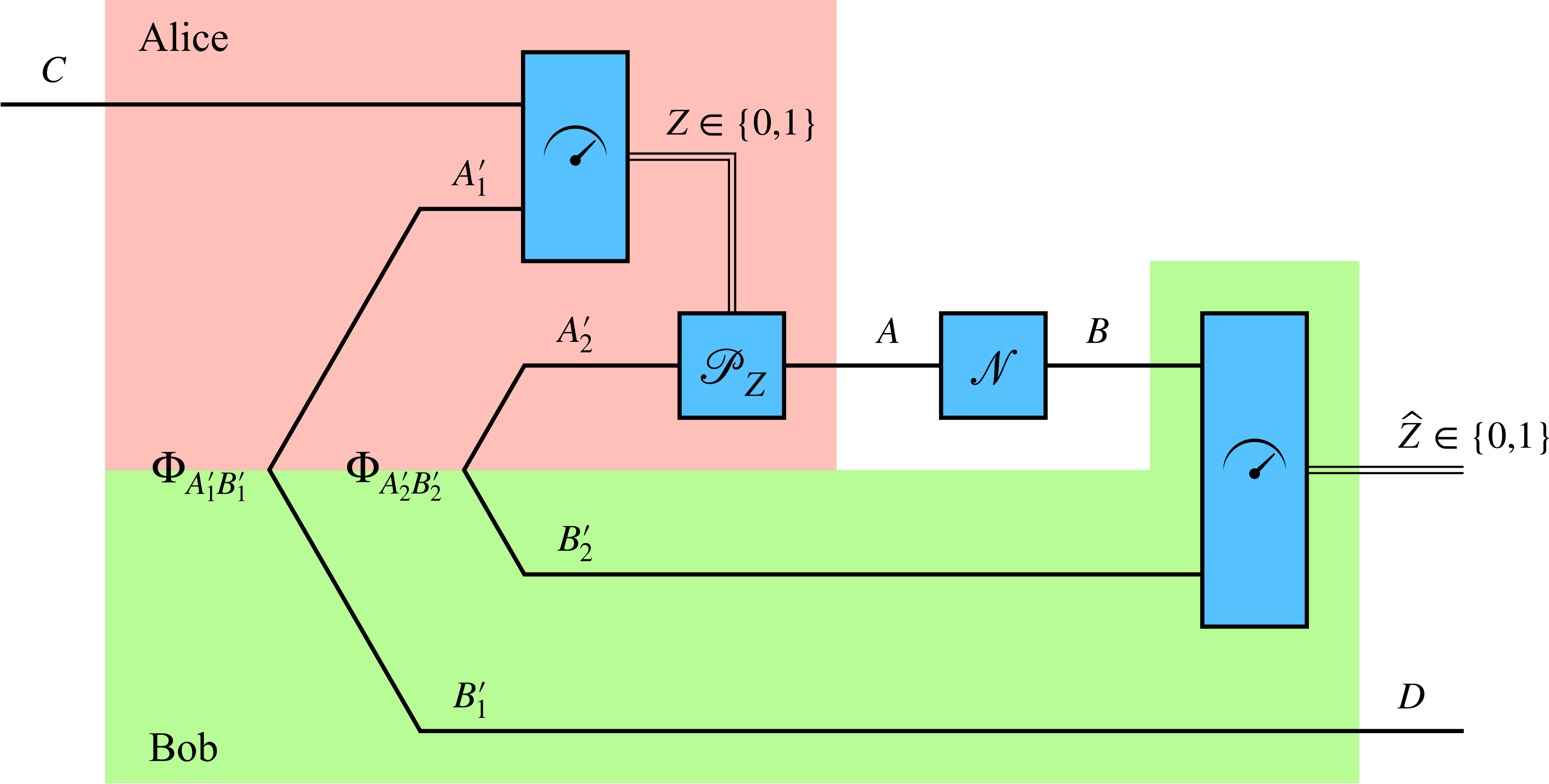
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Proof sketch: Achievability

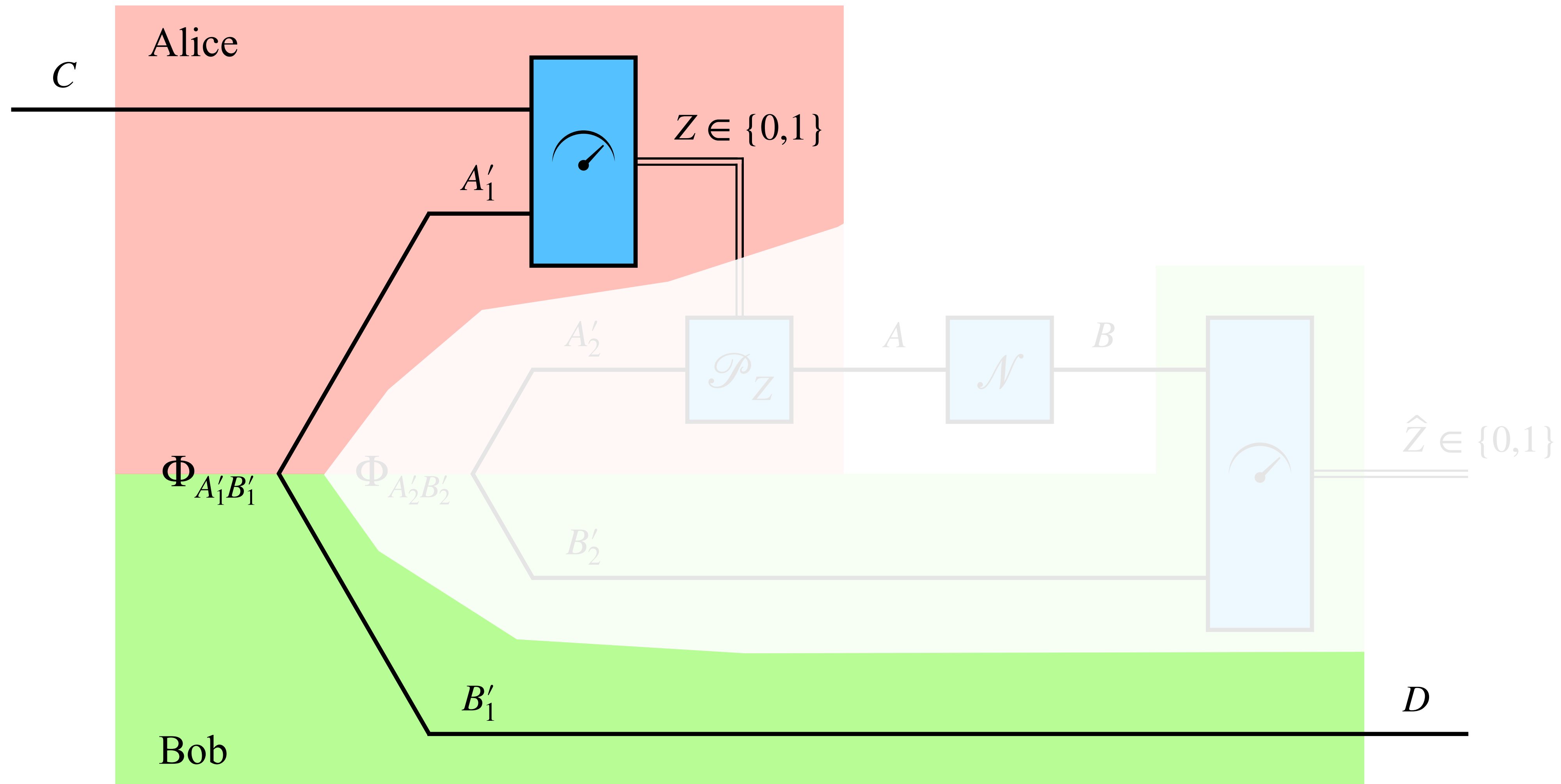
- Step 1: Simulate an “approximate **quantum** erasure channel”
 - Postselected teleportation-based coding
- Step 2: Simulate an “approximate **classical** erasure channel”
 - Super-dense coding

$$[q \rightarrow q] + [qq] \geq 2[c \rightarrow c] .$$

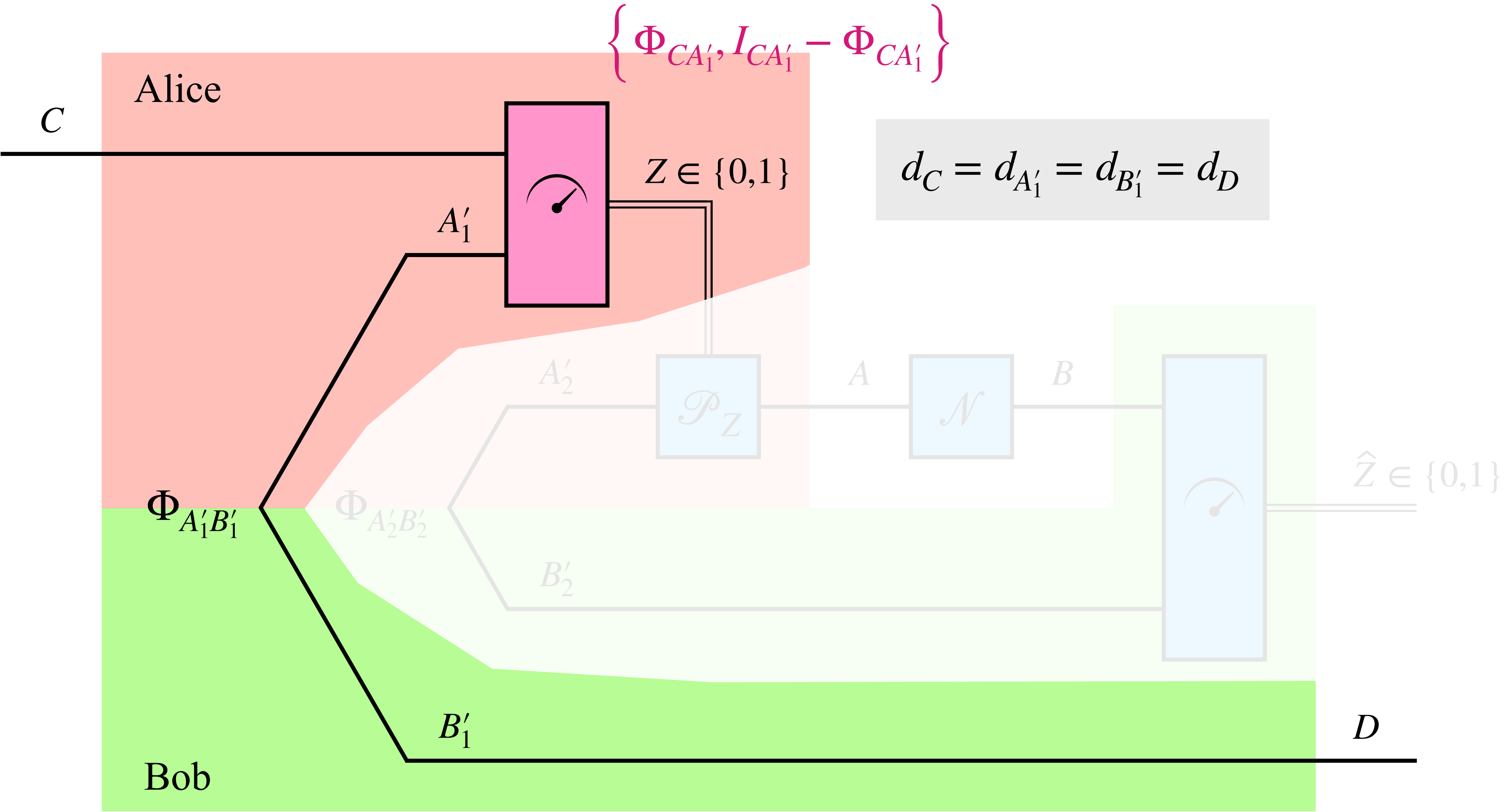
Step 1: Postselected teleportation-based coding



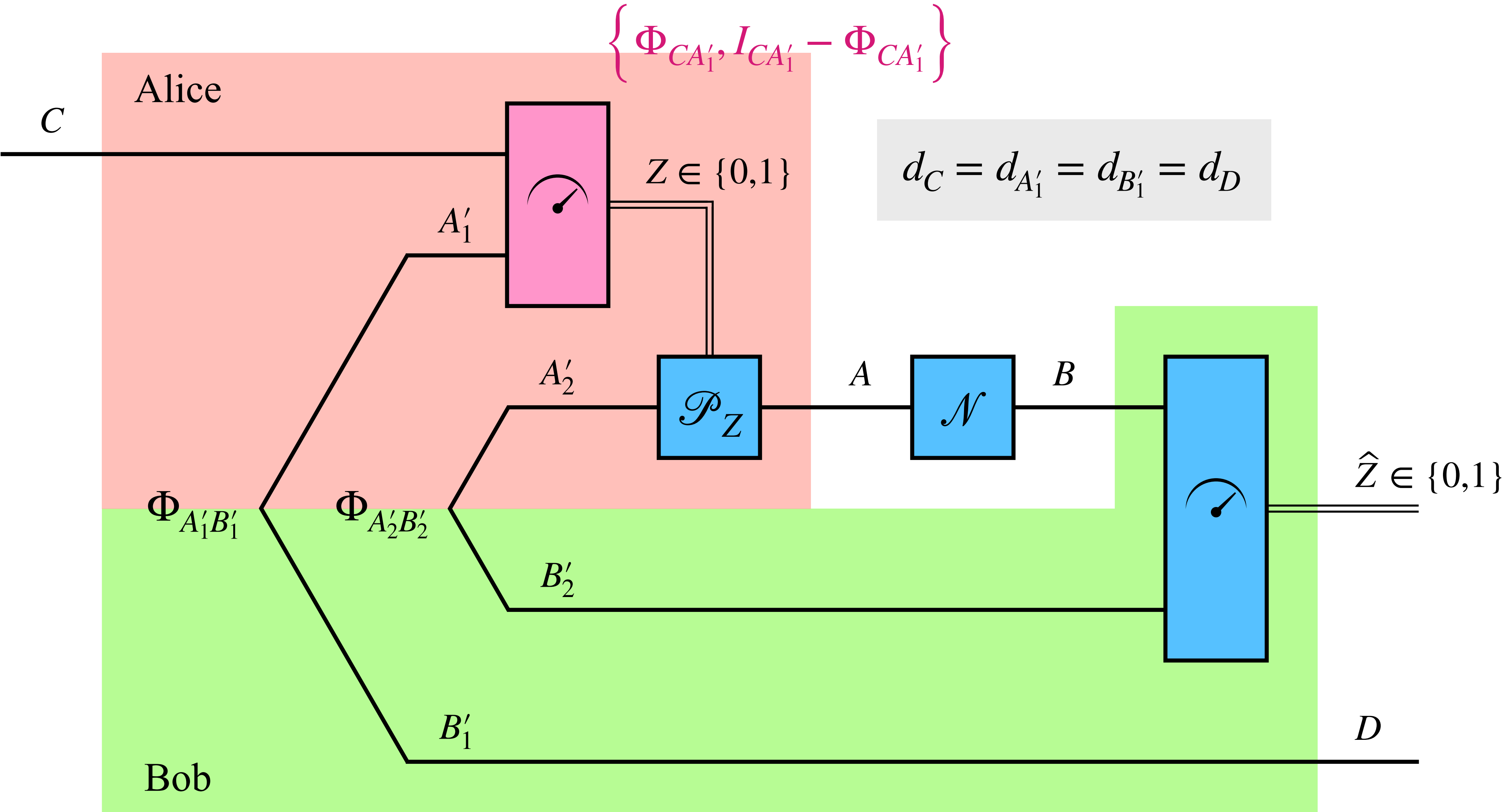
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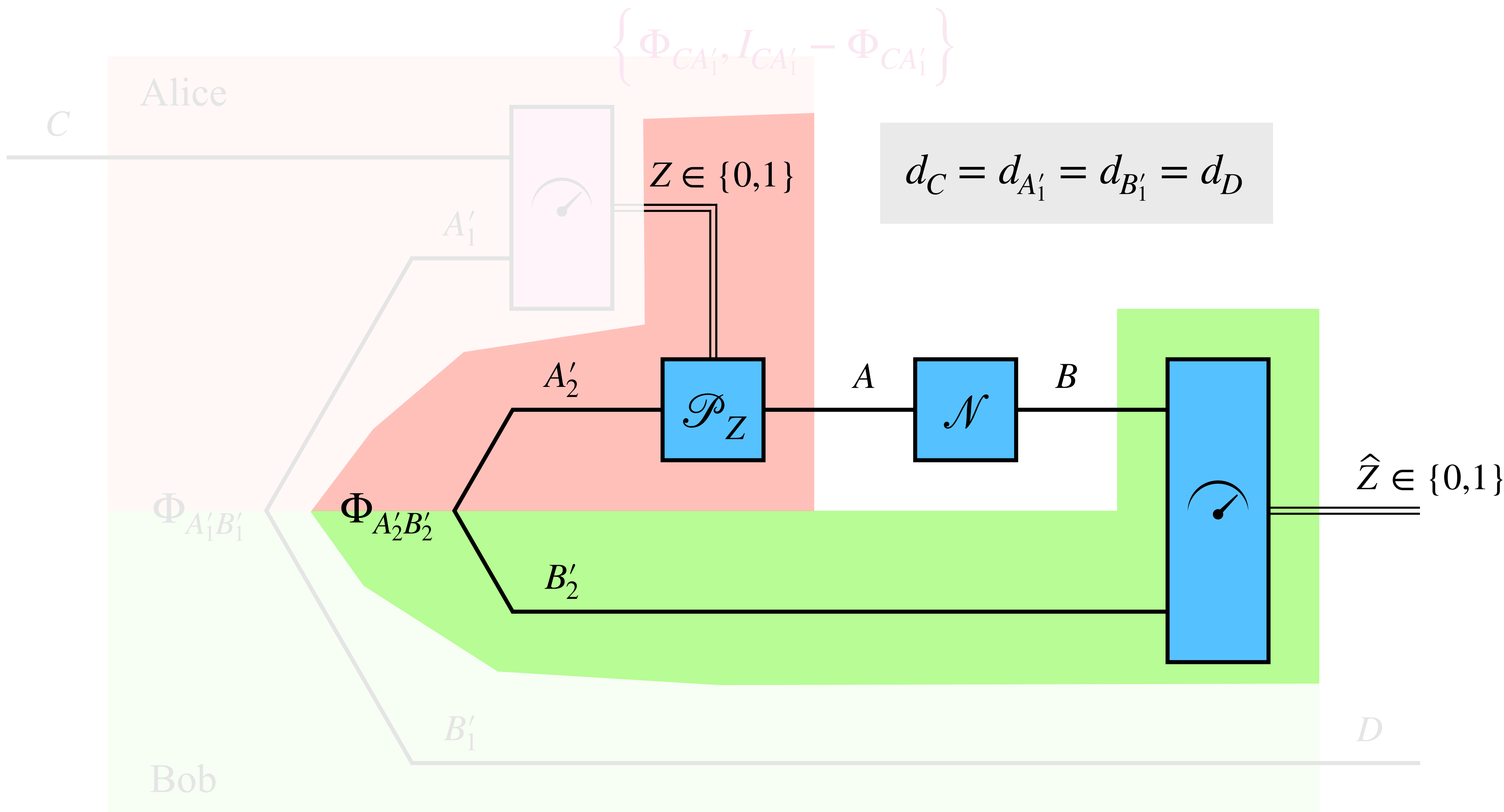
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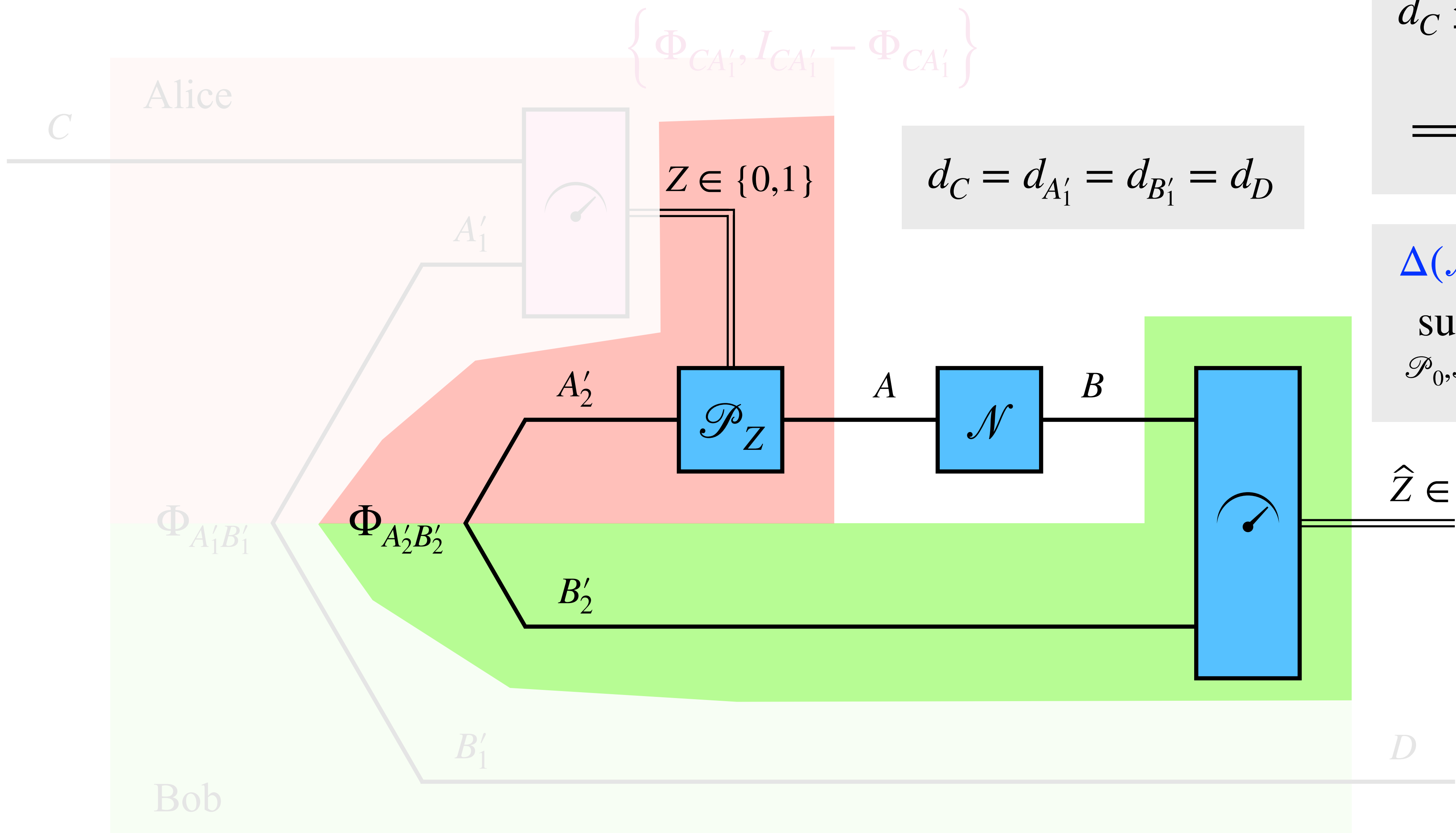
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$$d_C = d_{A_1} = d_{B_1} = d_D$$

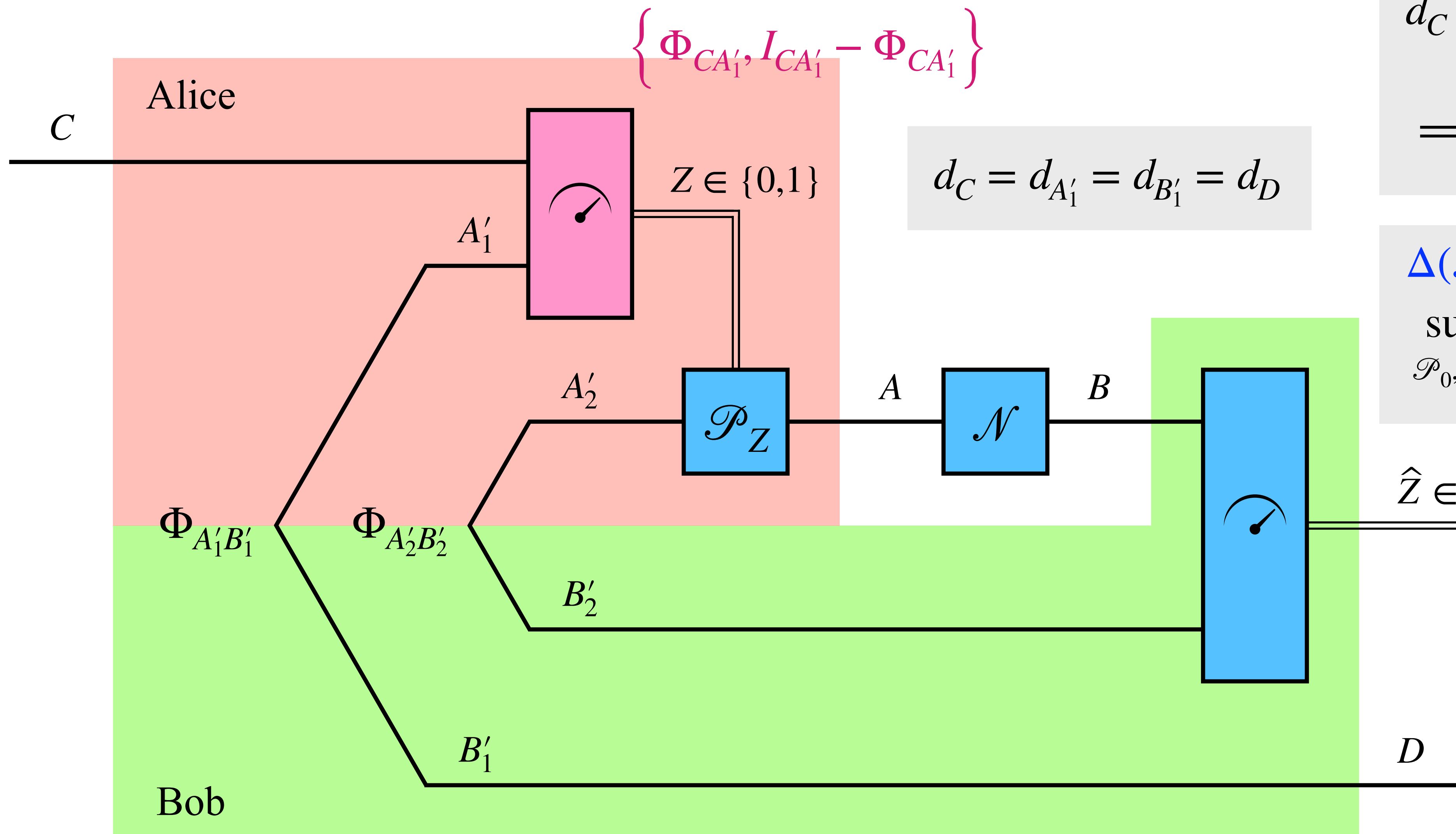
$$d_C \leq \sqrt{\frac{\varepsilon}{1-\varepsilon} 2^{\Delta(\mathcal{N})} + 1}$$

$$\implies \Pr \left\{ Z = 1 \mid \hat{Z} = 0 \right\} \leq \varepsilon$$

$$\Delta(\mathcal{N}) := \sup_{\mathcal{P}_0, \mathcal{P}_1} D_{\max}(\mathcal{N} \circ \mathcal{P}_0 \| \mathcal{N} \circ \mathcal{P}_1)$$

$$\hat{Z} \in \{0,1\}$$

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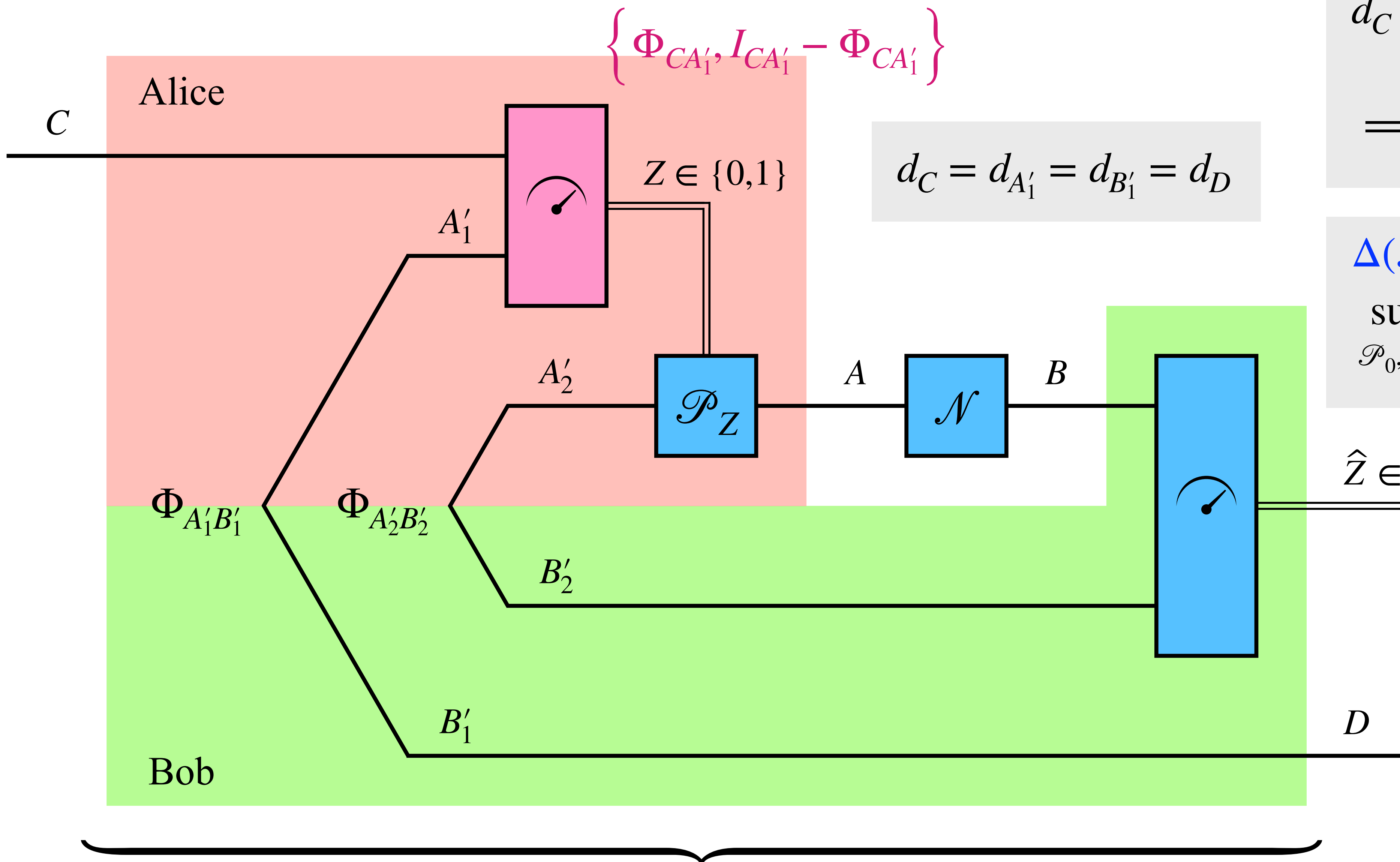


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$$\left\{ \Phi_{CA'_1}, I_{CA'_1} - \Phi_{CA'_1} \right\}$$

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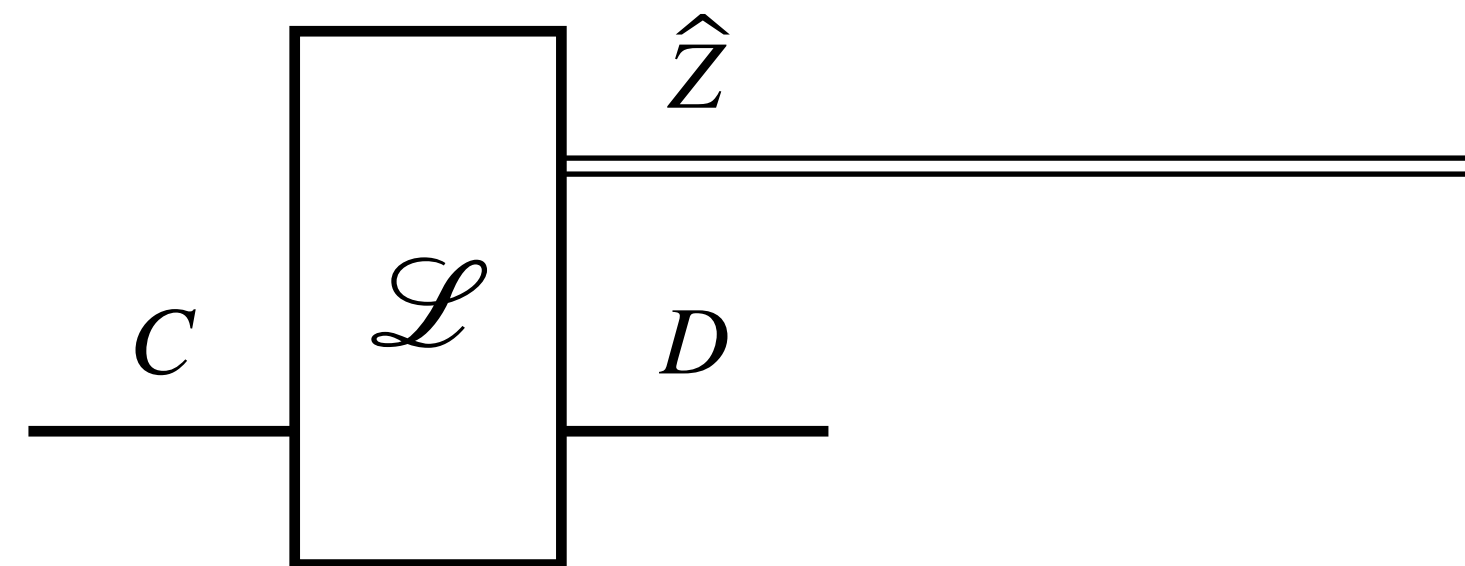
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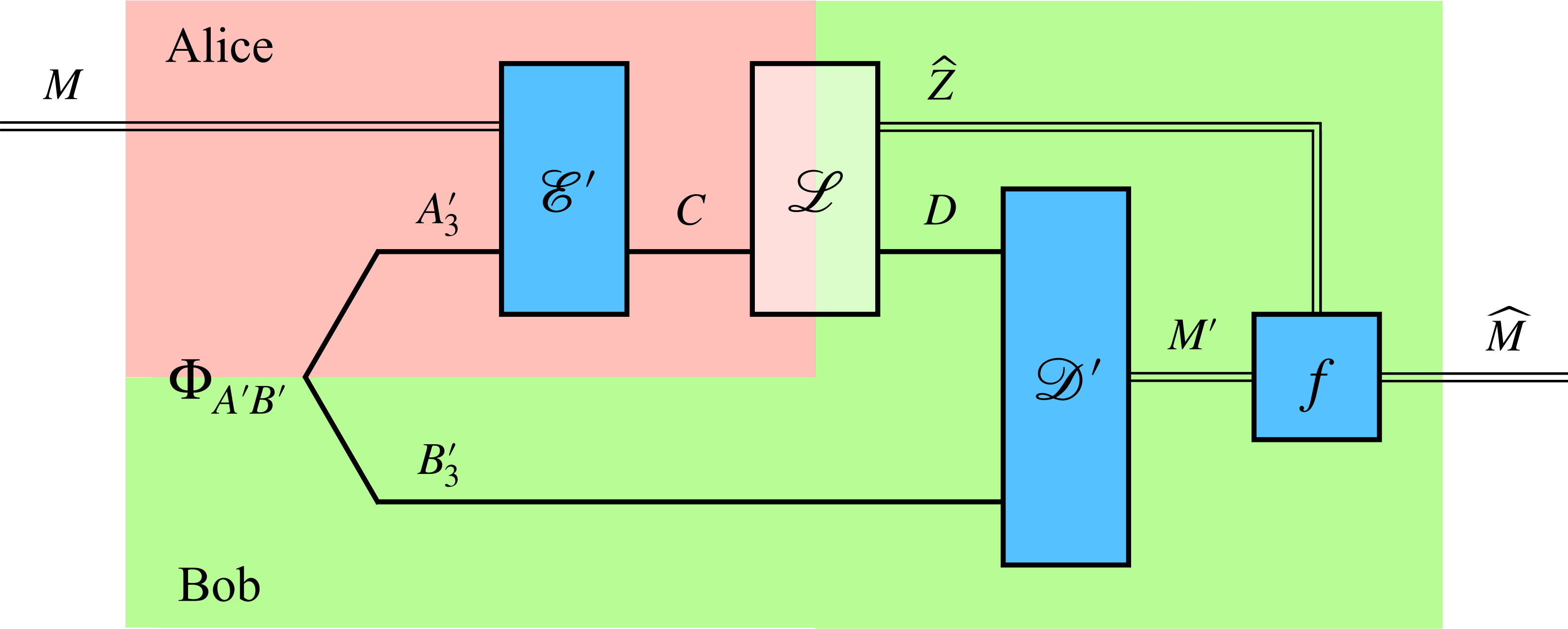
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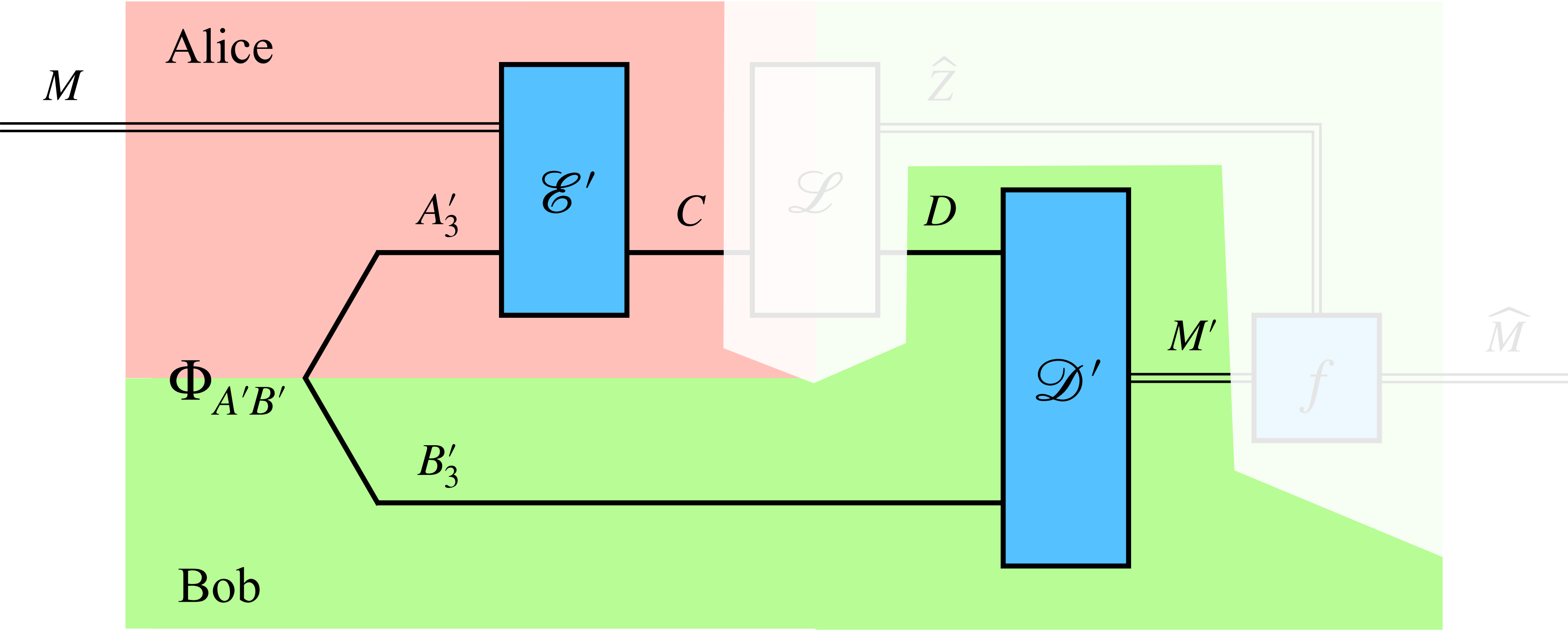
Step 2: Super-dense coding



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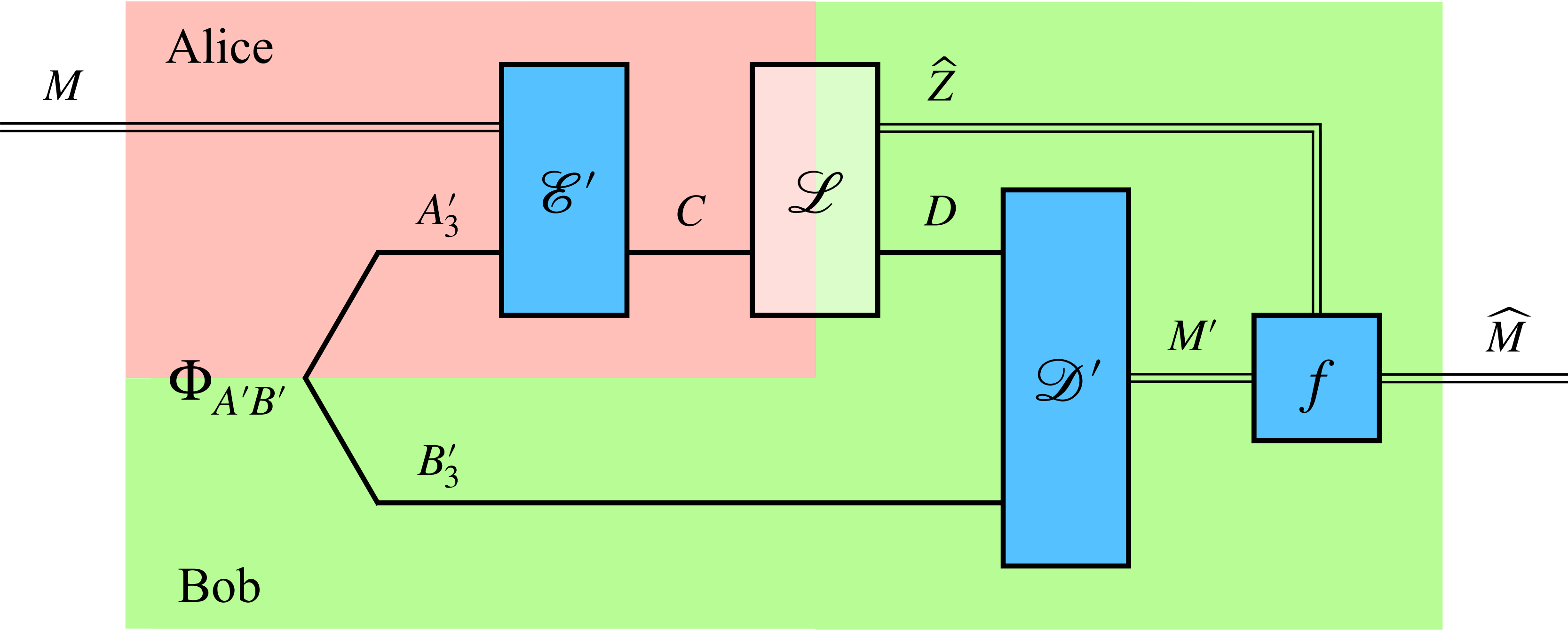
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$\mathcal{E}'$ and $\mathcal{D}'$ are the standard encoding and decoding operations for super-dense coding

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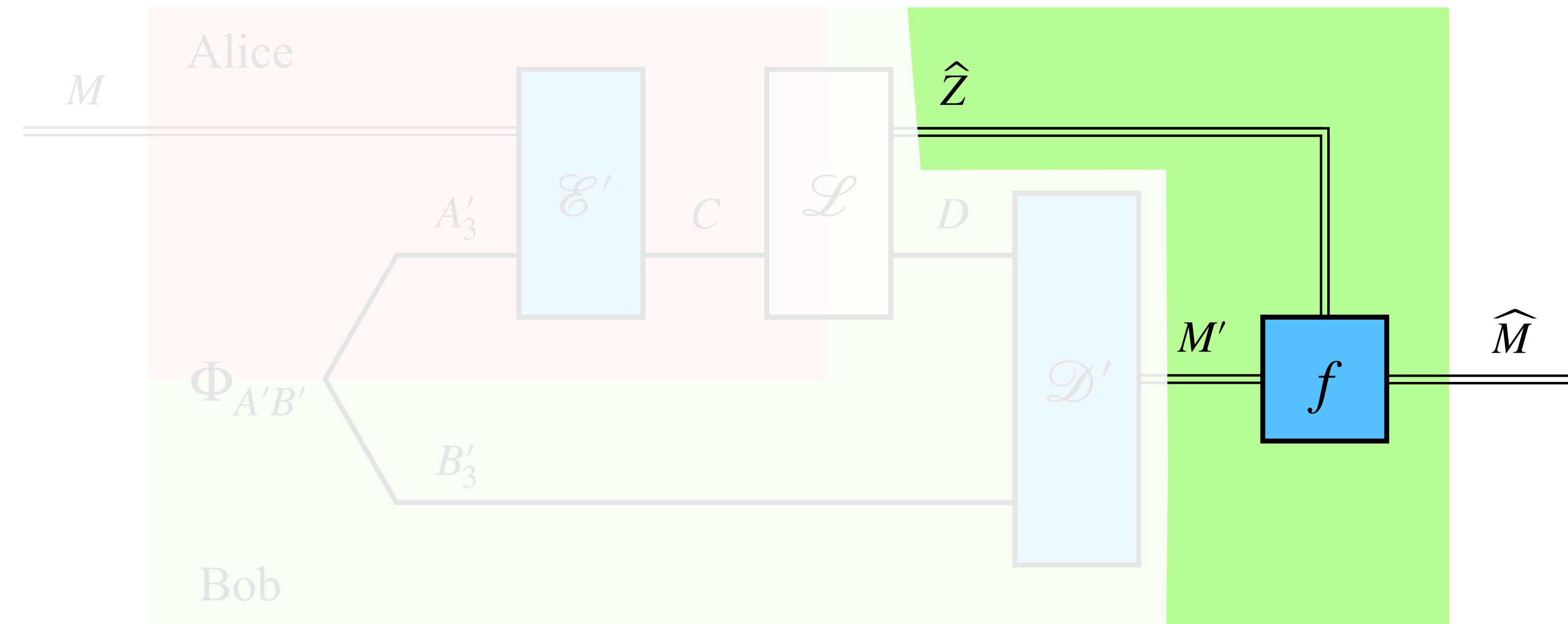
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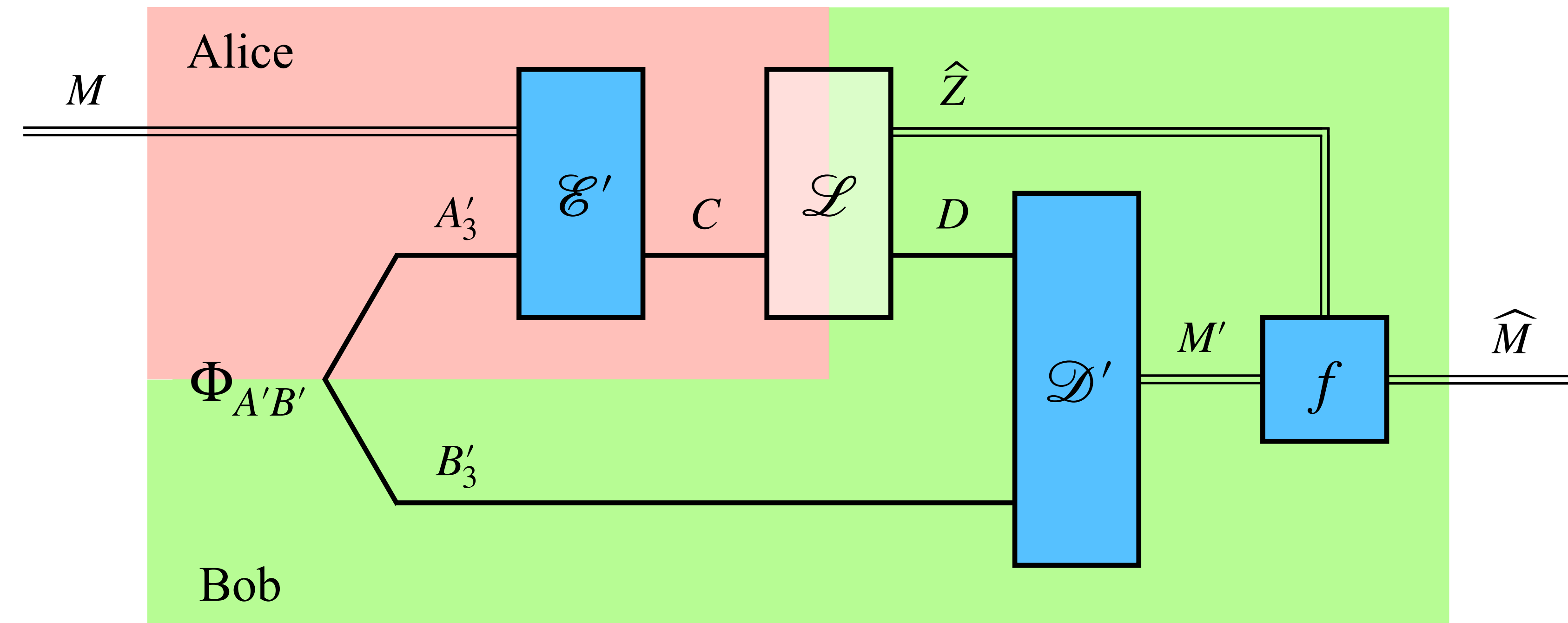
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Step 2: Super-dense coding



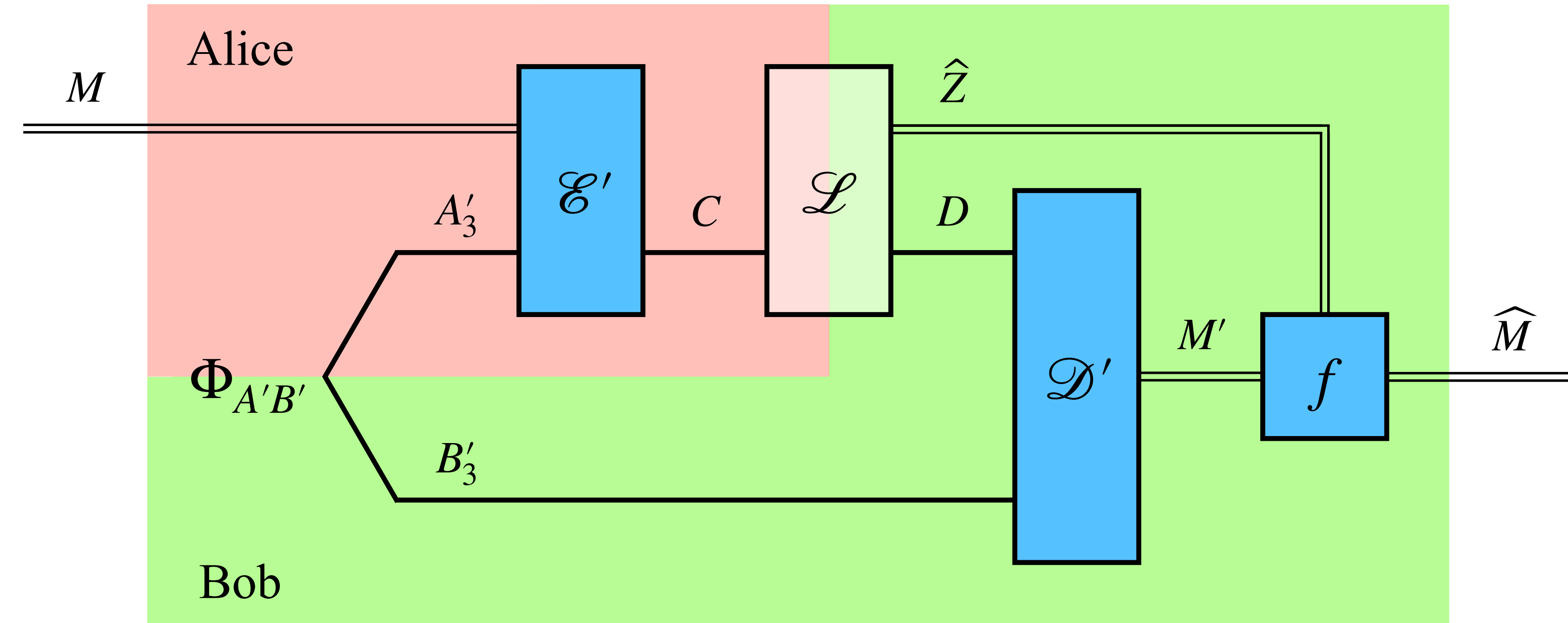
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Achievability bound (informal):

$$C_{\text{pEA}}^\varepsilon(\mathcal{N}) \gtrsim \log_2 \left(\frac{\varepsilon}{1-\varepsilon} 2^{\Delta(\mathcal{N})} + 1 \right).$$

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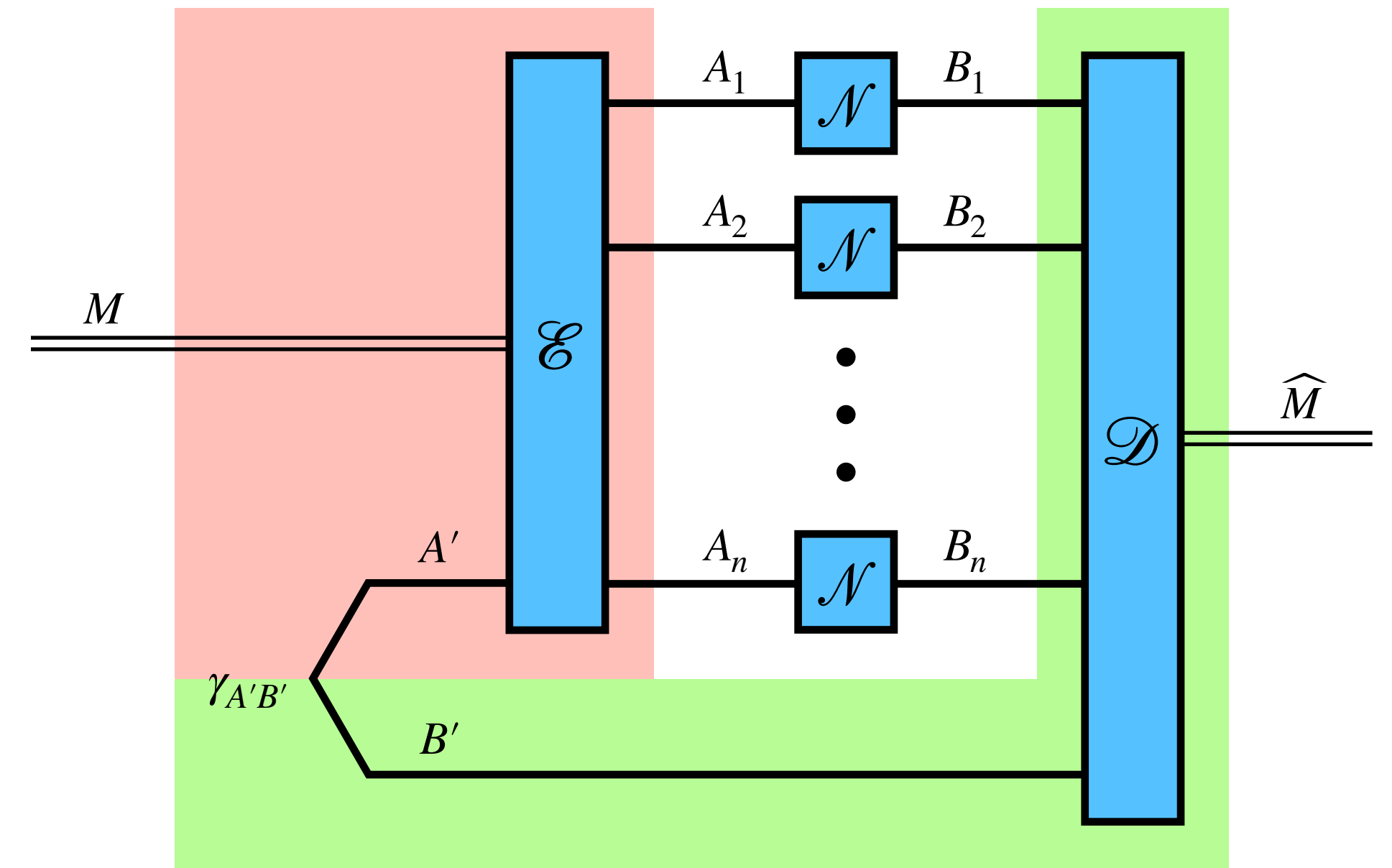
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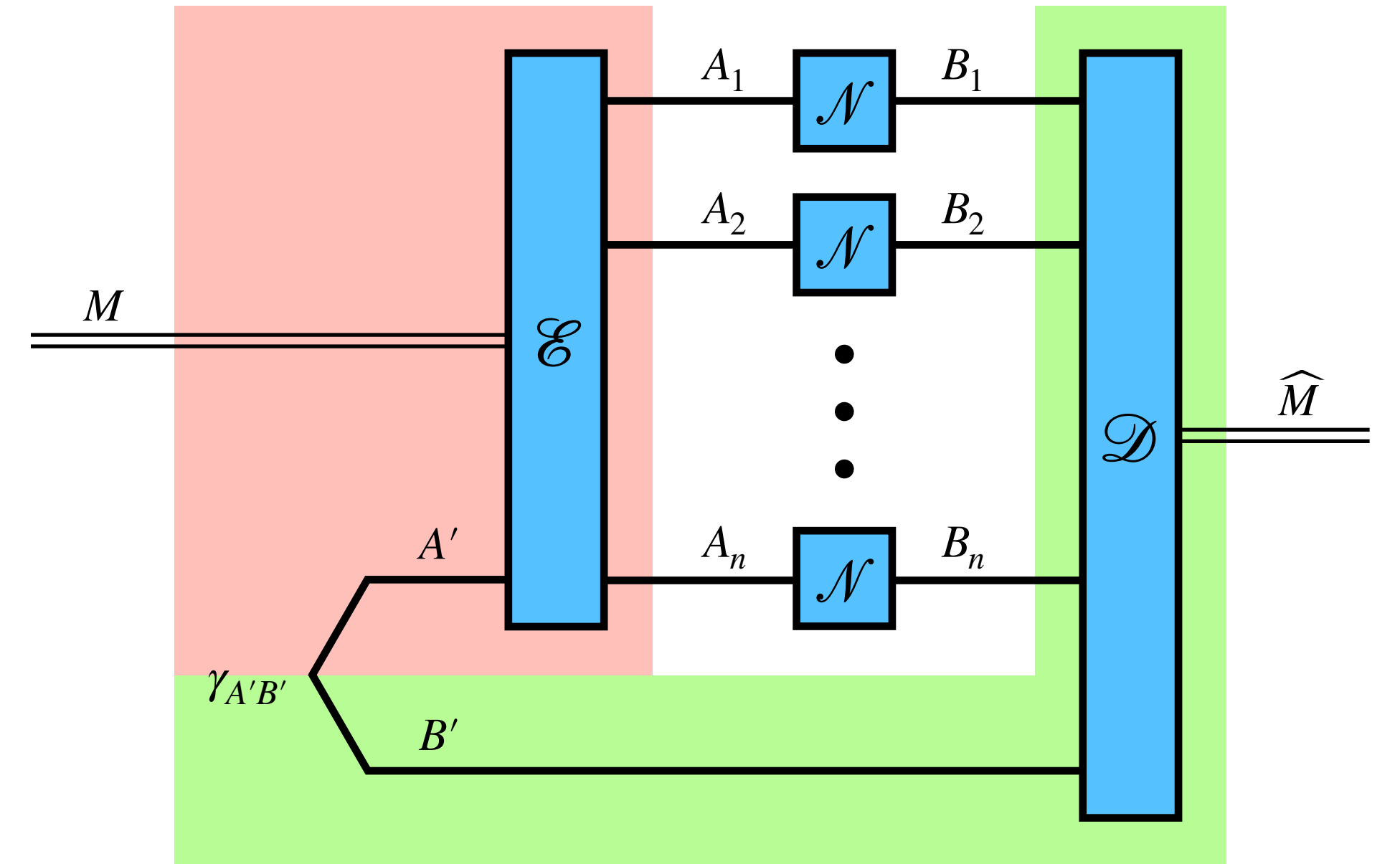
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Asymptotic pEA classical capacity



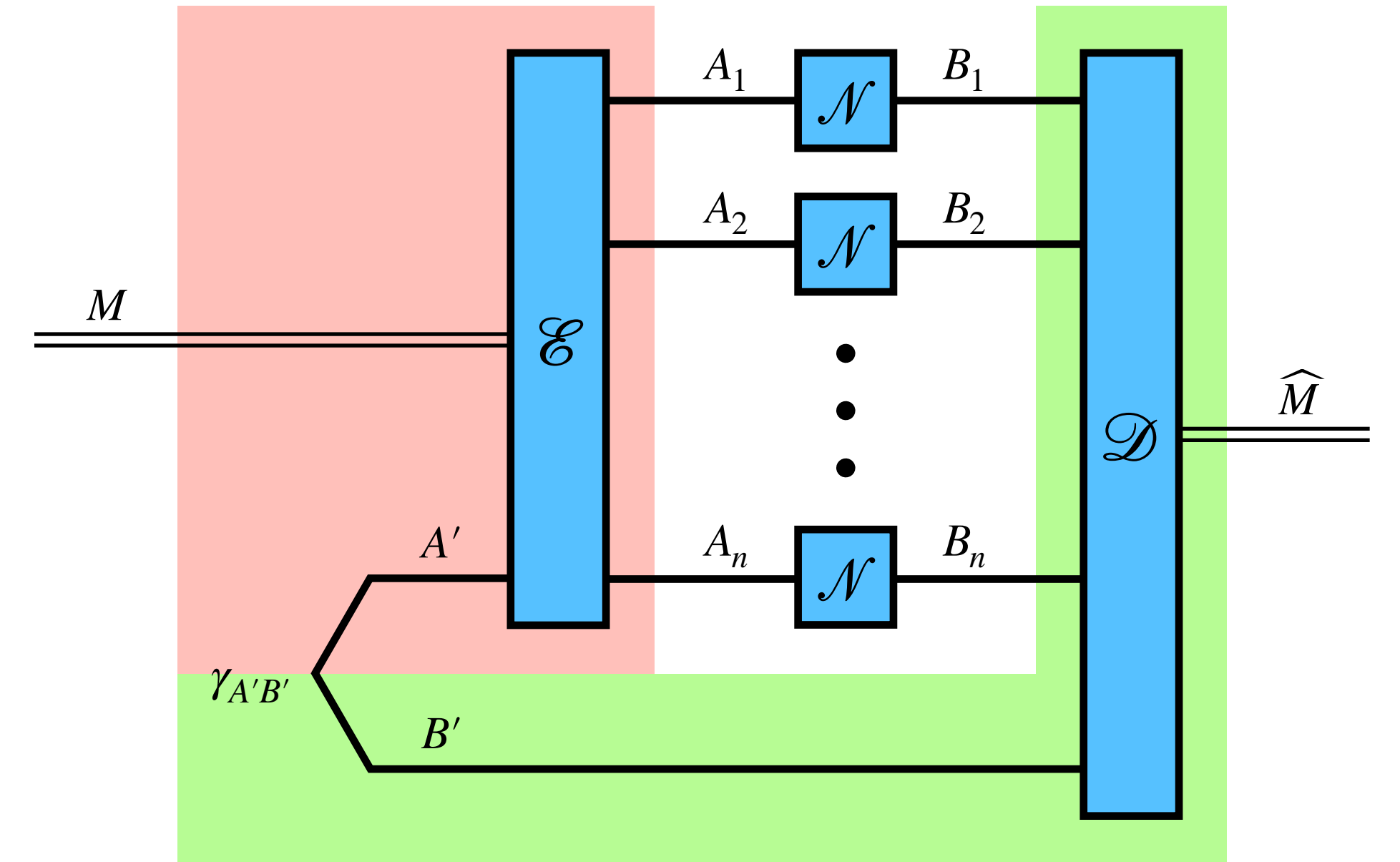
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$$\begin{aligned}
 C_{\text{pEA}}(\mathcal{N}) &:= \inf_{\varepsilon \in (0,1)} \liminf_{n \rightarrow \infty} \frac{1}{n} C_{\text{pEA}}^{\varepsilon}(\mathcal{N}^{\otimes n}) \\
 &= \inf_{\varepsilon \in (0,1)} \liminf_{n \rightarrow \infty} \frac{1}{n} \log_2 \left(\frac{\varepsilon}{1-\varepsilon} 2^{I_{\Omega}(\mathcal{N}^{\otimes n})} + 1 \right) \\
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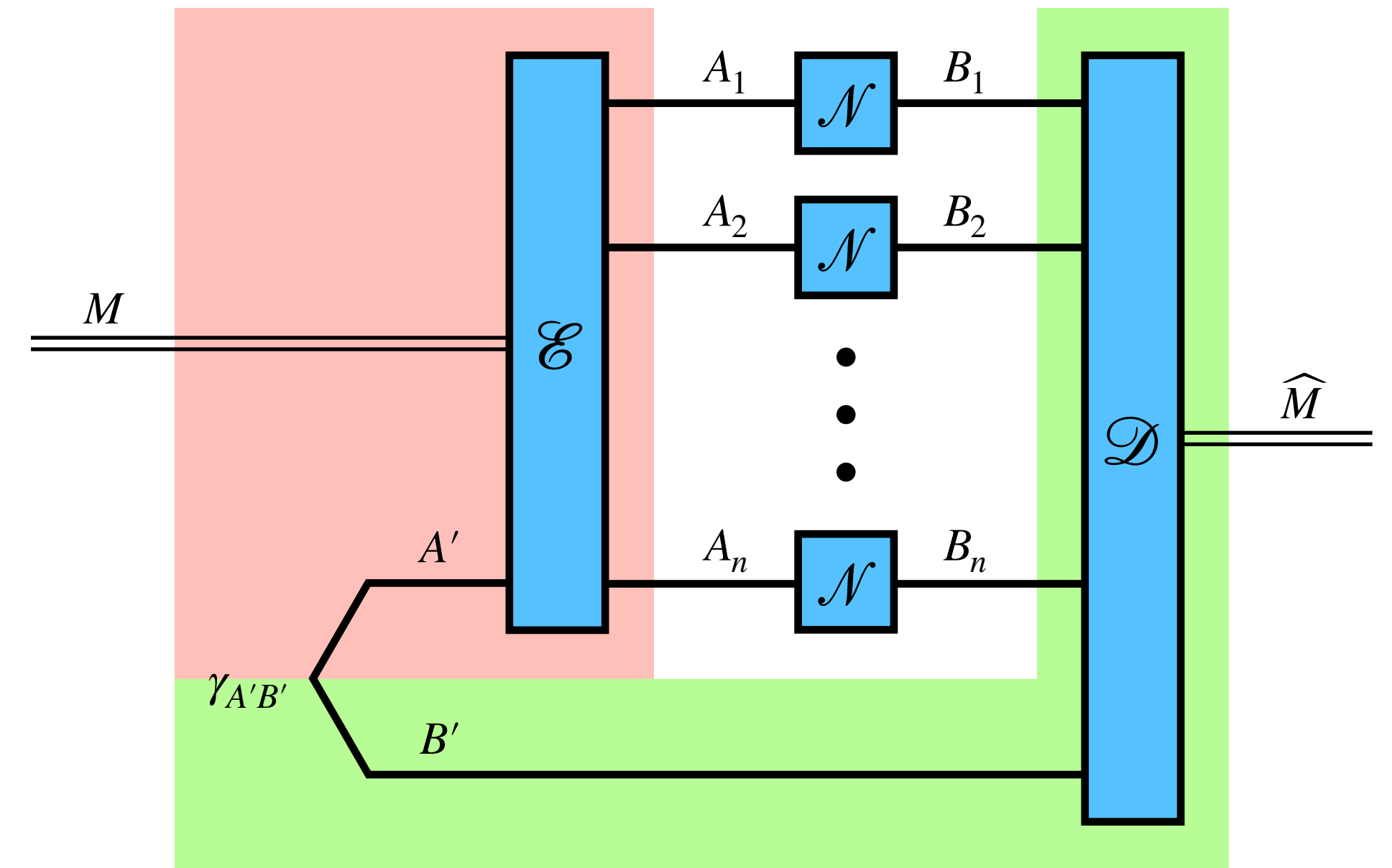


Additivity of $I_{\Omega}(\mathcal{N})$:

$$I_{\Omega}(\mathcal{N} \otimes \mathcal{M}) = I_{\Omega}(\mathcal{N}) + I_{\Omega}(\mathcal{M}) .$$

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Asymptotic pEA classical capacity:

$$C_{\text{pEA}}(\mathcal{N}) = \widetilde{C}_{\text{pEA}}(\mathcal{N}) = I_{\Omega}(\mathcal{N}) .$$

No second-order asymptotics.

Additional & corollary results

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 - One half of classical capacities (one-shot bounds & asymptotic)
 - Using duality between teleportation and super-dense coding

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 - One half of classical capacities (one-shot bounds & asymptotic)
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- Postselected nonsignalling-assisted (pNA) capacities
 - The same as pEA capacities (one-shot bounds & asymptotic)

Additional & corollary results

- pEA quantum capacities
 - One half of classical capacities (one-shot bounds & asymptotic)
 - Using duality between teleportation and super-dense coding
- Postselected nonsignalling-assisted (pNA) capacities
 - The same as pEA capacities (one-shot bounds & asymptotic)
- Feedback assistance
 - Provides zero advantage, even if non-asymptotic

Summary

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 - Decoder is granted an additional option of “making no guess” (i.e., being inconclusive); inconclusive trials are dismissed
 - Definitions of error probability and capacities change accordingly
- Characterisation of one-shot & asymptotic, pEA & pNA, classical & quantum capacities and their strong converses:
 - $C_{\text{pEA}}^\varepsilon(\mathcal{N}) \approx C_{\text{pNA}}^\varepsilon(\mathcal{N}) \approx 2Q_{\text{pEA}}^\varepsilon(\mathcal{N}) \approx 2Q_{\text{pNA}}^\varepsilon(\mathcal{N}) \approx \log_2 \left(\frac{\varepsilon}{1-\varepsilon} 2^{I_\Omega(\mathcal{N})} + 1 \right).$
 - $C_{\text{pEA}}(\mathcal{N}) = C_{\text{pNA}}(\mathcal{N}) = 2Q_{\text{pEA}}(\mathcal{N}) = 2Q_{\text{pNA}}(\mathcal{N})$
 $= \widetilde{C}_{\text{pEA}}(\mathcal{N}) = \widetilde{C}_{\text{pNA}}(\mathcal{N}) = 2\widetilde{Q}_{\text{pEA}}(\mathcal{N}) = 2\widetilde{Q}_{\text{pNA}}(\mathcal{N}) = I_\Omega(\mathcal{N}).$

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- Future work
 - Postselected unassisted communication
 - Tradeoff between postselected rate and conclusive probability